

吉林大学

DB2 实验报告

班级：计科 4 班

姓名：马兆星

学号：21230415

姓名	马兆星	学号	21230415
实验项目	3. Create objects		
实验性质	☐演示性实验☐验证性实验 ☒操作性实验☐综合性实验		
实验地点	计楼A210	机器编号	
指导教师	康辉	实验时间	2025 年 11 月 23 日 8 时 30 分

一、实验综述

1. 实验目的及要求

This exercise is an online lab which creates objects used in a database.

At the end of the lab, students should be able to:

- Create tables
- Add referential integrity constraints to a table
- Create indexes
- Add check constraints to a table
- Create views
- Add triggers to a table
- Create an alias
- Access System Catalog information about objects
- Retrieve an XML document

2. 实验设备、软件

PC, windows XP Professional, DB2 9 Express-c

二、实验过程（实验步骤、记录、数据、分析）

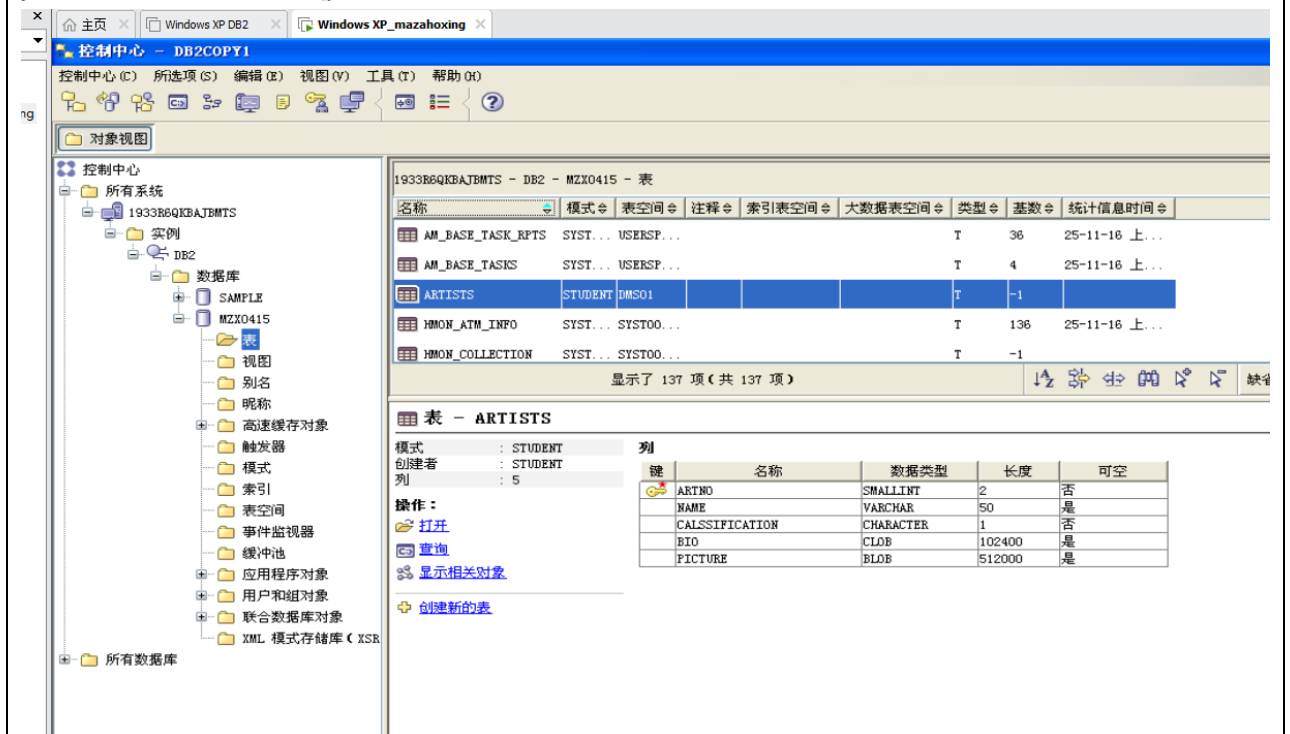
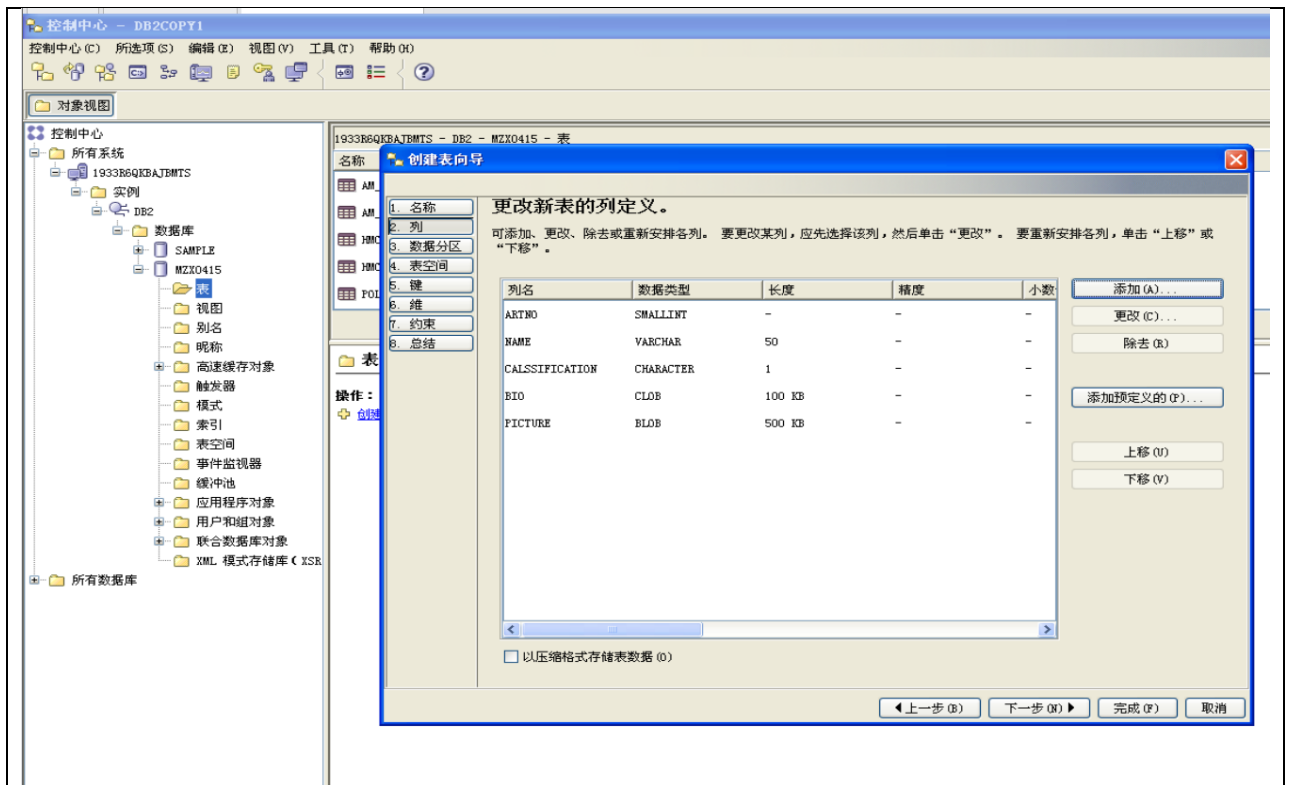
Section 1 - Create Tables

1. Create your ARTISTS table. The table should have the following characteristics.

- Table schema of student
- Table Name of artists
- Regular data to go in table space dms01
- Indexes to go in table space dms02
- Long data to go in table space dms03
- The column artno should be defined as a primary key
- Columns defined like the following:

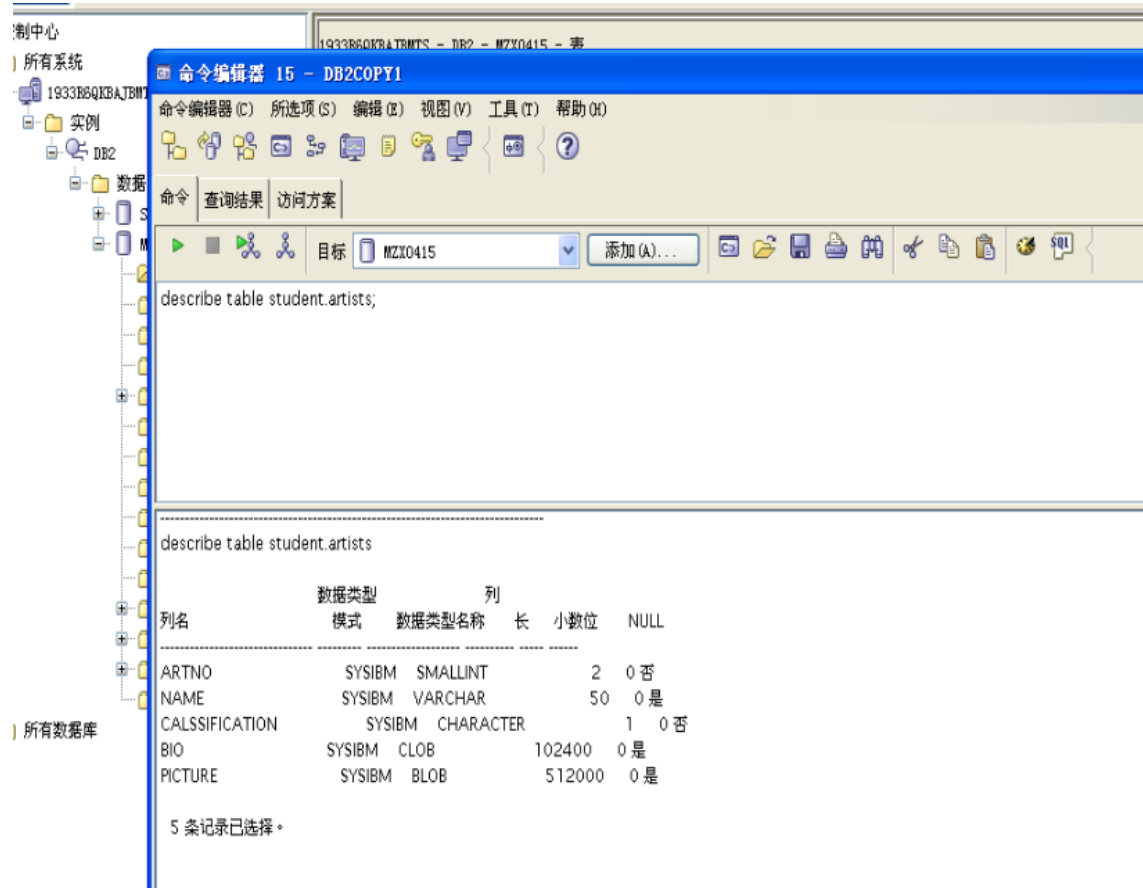
(artno smallint not null,
name varchar (50),
classification char (1) not null,
bio clob (100K) logged compact,
picture blob (500k) not logged compact)

Show your work below by printing screen.



2. Verify the creation of your ARTISTS table and the table description.
Important: Ensure the definition of your table is correct. It could cause unexpected failures in later labs if your definitions are not done correctly.

Show your work below by printing screen.



3. You will create the remainder of the tables required using a script file called crtables. For Windows, it is located in the C:\labfiles\cf23 directory.

Examine the DB2 statements in the file and answer the following questions.

In which table space will both the indexes and data for table STOCK be placed? For CONCERTS? For REORDER?

4. Execute the crtables script to create the tables.

Show your work below by printing screen.

create table albums
(title varchar(50),
artno smallint not null,
itemno smallint not null)
in dms04
index in dms05;

create table stock
(itemno smallint not null,
type char(1) not null,

create table albums (title varchar(50), artno smallint not null, itemno smallint not null) in dms04 index in dms05
DB20000I SQL 命令成功完成。

create table stock (itemno smallint not null, type char(1) not null, price decimal(5,2) not null with default, qty int not null with default) in dms06
DB20000I SQL 命令成功完成。

create table concerts (artno smallint not null, date date not null, city varchar(25) not null with default) in dms04
DB20000I SQL 命令成功完成。

create table reorder (itemno smallint not null, timestamp timestamp) in sms01
DB20000I SQL 命令成功完成。

与目标的 JDBC 连接已成功。

语句终止字符

控制中心 (C) 所选项 (S) 编辑 (E) 视图 (V) 工具 (T) 帮助 (H)

对象视图

控制中心

所有系统

1933R6QKBAJMTS

实例

DB2

数据库

SAMPLE

MZX0415

表

视图

别名

昵称

高速缓存对象

触发器

模式

索引

表空间

事件监视器

缓冲池

应用程序对象

用户和组对象

联合数据库对象

XML 模式存储库 (XSR)

所有数据库

1933R6QKBAJMTS - DB2 - MZX0415 - 表

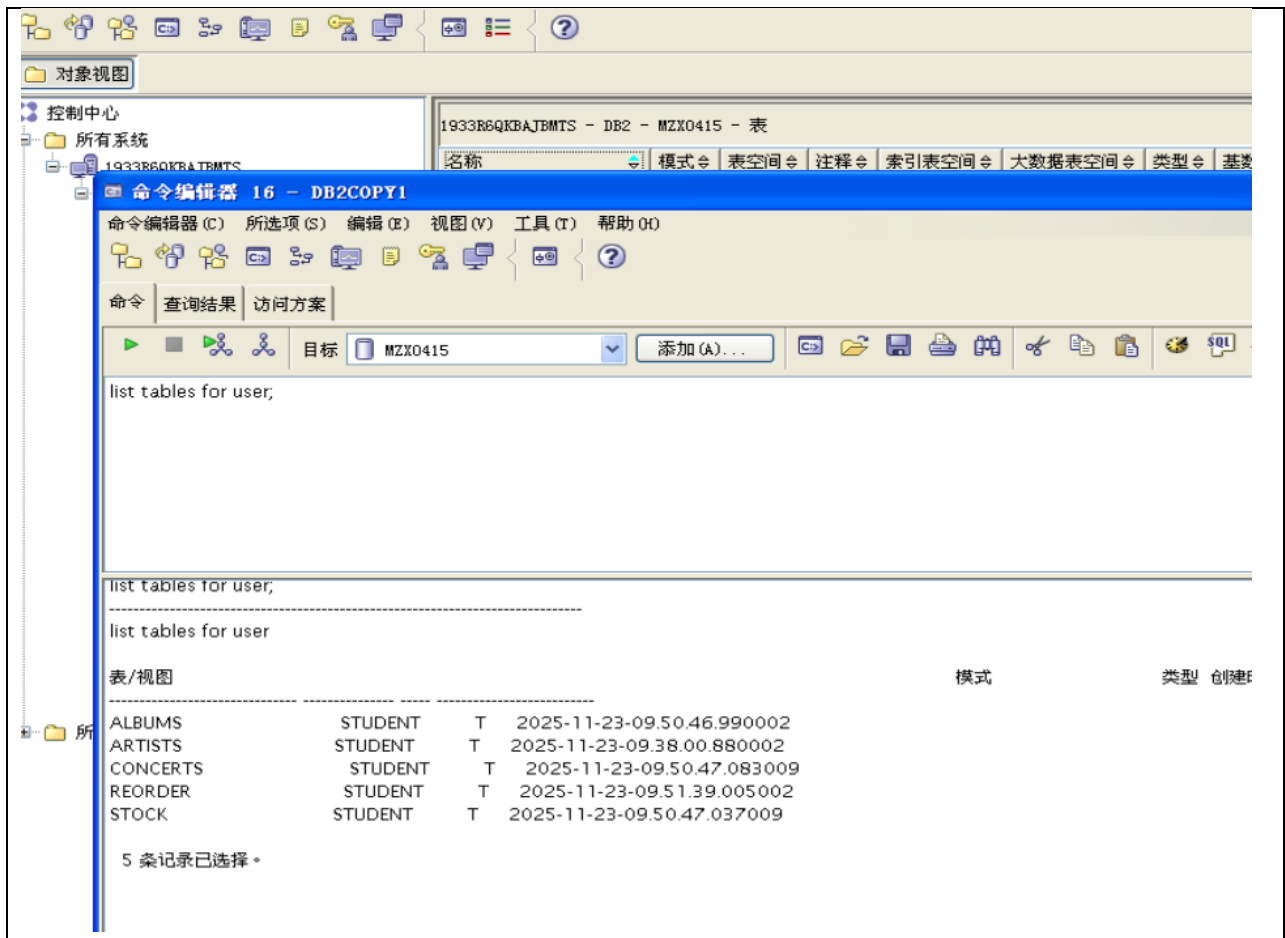
名称	模式	表空间	注释	索引表空间	大数据表空间	类型	基数	统计信息时间
ALBUMS	STUDENT	DMS04		DMS05		T	-1	
AM_BASE_TASK_RPTS	SYST...	USERSP...				T	36	25-11-16 上...
AM_BASE_TASKS	SYST...	USERSP...				T	4	25-11-16 上...
ARTISTS	STUDENT	DMS01				T	0	25-11-23 上...
CONCERTS	STUDENT	DMS04				T	-1	
HMON_ATM_INFO	SYST...	SYST00...				T	136	25-11-16 上...
HMON_COLLECTION	SYST...	SYST00...				T	-1	
POLICY	SYST...	SYST00...				T	5	25-11-16 上...
REORDER	STUDENT	SMS01				T	-1	
STMG_DBSIZE_INFO	SYST...	SYST00...				T	1	25-11-16 上...
STOCK	STUDENT	DMS06				T	-1	
SYSATTRIBUTES	SYSIBM	SYSCAT...				T	-1	
SYSAUDITPOLICIES	SYSIBM	SYSCAT...				T	0	25-11-16 上...
SYSAUDITUSE	SYSIBM	SYSCAT...				T	-1	
SYSBUFFERPOOLNODES	SYSIBM	SYSCAT...				T	-1	
SYSBUFFERPOOLS	SYSIBM	SYSCAT...				T	1	25-11-23 上...
SYSCHECKS	SYSIBM	SYSCAT...				T	1	25-11-16 上...
SYSCODEPROPERTIES	SYSIBM	SYSCAT...				T	-1	
SYSCOLAUTH	SYSIBM	SYSCAT...				T	0	25-11-16 上...
SYSCOLCHECKS	SYSIBM	SYSCAT...				T	1	25-11-16 上...
SYSCOLDIST	SYSIBM	SYSCAT...				T	-1	

显示了 141 项 (共 141 项)

表 - HMON_ATM_INFO

5. Verify that all tables (ALBUMS, STOCK, CONCERTS, and REORDER) were created successfully.

Show your work below by printing screen.



6. Information about each column in a table can be found by accessing the SYSCAT.COLUMNS view.

Enter an interactive SQL statement to select the columns TABNAME, COLNAME, and TYPENAME from SYSCAT.COLUMNS for all tables with the TABSCHEMA of 'student', order the data by TABNAME and COLNO, and pipe the output to a file named tabchk.file.

Show your sql statements and your work below by printing screen for the last step.

TABNAME	COLNAME
ALBUMS	ARTNO
ALBUMS	ITEMNO
ALBUMS	TITLE
ARTISTS	ARTNO
ARTISTS	BIO
ARTISTS	CLASSIFICATION
ARTISTS	NAME
ARTISTS	PICTURE
CONCERTS	ARTNO
CONCERTS	CITY
CONCERTS	DATE
REORDER	ITEMNO
REORDER	TIMESTAMP
STOCK	ITEMNO
STOCK	PRICE
STOCK	QTY
STOCK	TYPE

17 条记录已选择。

7. Check your output against the tabchk.master file.

Ensure that there are no differences between your output and the master output or the following labs may not work correctly!

Show your work below by printing screen.

The screenshot shows a DB2 CLP terminal window with the following content:

```

SELECT TABNAME, COLNAME, TYPENAME
FROM SYSCAT.COLUMNS
WHERE TABSCHEMA = 'STUDENT' -- 匹配实验要求的表模式 (注意大写, DB2 对标识符大小写敏感)
ORDER BY TABNAME, COLNO -- 按表名、列序号排序, 确保结果有序
> tabchk.file -- 将输出重定向到 tabchk.file 文件 (Windows 环境适用)

D:\Labfiles\CF23>-tf tabchk.sql > tabchk.file
'-tf' 不是内部或外部命令, 也不是可运行的程序
或批处理文件。

D:\Labfiles\CF23>db2 -tf tabchk.sql > tabchk.file
SQL0104N

D:\Labfiles\CF23>fc /w tabchk.file tabchk.master : more
正在比较文件 tabchk.file 和 TABCHK.MASTER
***** tabchk.file
ARTISTS
说明:
CALSSIFICATION
CHARACTER
***** TABCHK.MASTER
ARTISTS
CLASSIFICATION
CHARACTER
用户响应:
*****
在指定的标记区域内检查并更正语句。
  
```

On the right side of the terminal window, there is a note: 注意大写, DB2 对标识符大小写敏感 (Note uppercase, DB2 is case-sensitive for identifiers).

8. Information about a table's definition to a table space can be found by accessing the SYSCAT.TABLES view. Enter an SQL statement to select the columns TABNAME, TBSPACE, and INDEX_TBSPACE from SYSCAT.TABLES for all tables with the TABSCHEMA of 'student', order the data by TABNAME, and pipe the output to a file named tbschk.file.

Show your sql statements and your work below by printing screen for the last step.



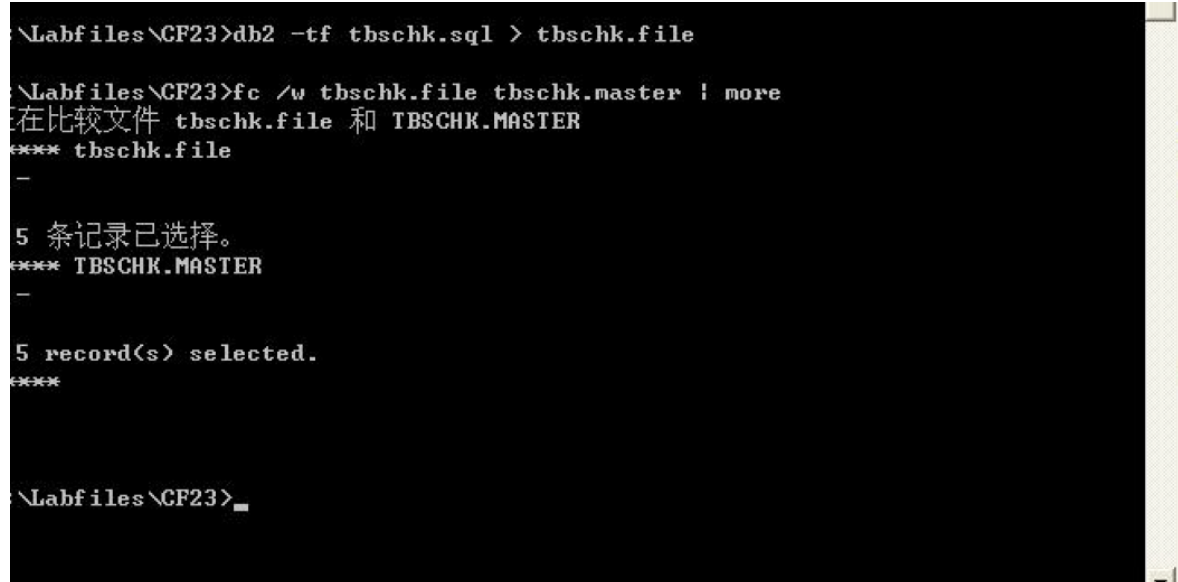
TABNAME	TBSPACE
ALBUMS	DMS 04
ARTISTS	DMS 01
COMMENTS	DMS 04
REORDER	SMS 01
STOCK	DMS 06

5 条记录已选择。

9. Check your output against the tbschk.master file.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.

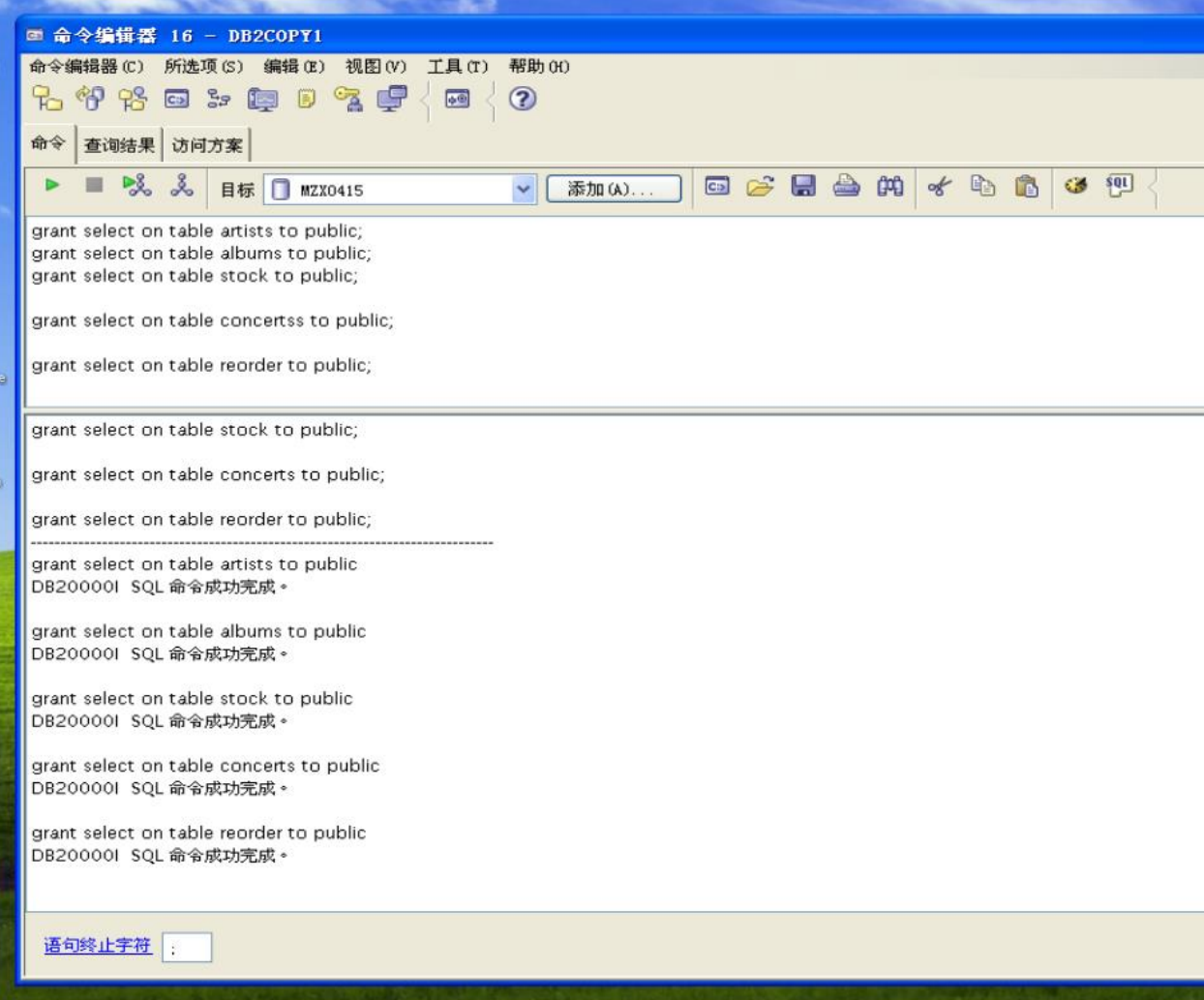


```
\Labfiles\CF23>db2 -tf tbschk.sql > tbschk.file

\Labfiles\CF23>fc /u tbschk.file tbschk.master ! more
在比较文件 tbschk.file 和 TBSCHK.MASTER
**** tbschk.file
-
5 条记录已选择。
**** TBSCHK.MASTER
-
5 record(s) selected.
****

\Labfiles\CF23>
```

10. Grant select privilege on tables that are owned by student to public by executing a script file, grants (an operating system command file) for local administration.
Show your work below by printing screen.



```
命令编辑器 16 - DB2COPY1
命令编辑器(C) 所选项(S) 编辑(E) 视图(V) 工具(T) 帮助(H)

目标 MZX0415 添加(A)...

grant select on table artists to public;
grant select on table albums to public;
grant select on table stock to public;

grant select on table concertss to public;

grant select on table reorder to public;

grant select on table stock to public;

grant select on table concerts to public;

grant select on table reorder to public;
-----
grant select on table artists to public
DB20000I SQL 命令成功完成。

grant select on table albums to public
DB20000I SQL 命令成功完成。

grant select on table stock to public
DB20000I SQL 命令成功完成。

grant select on table concerts to public
DB20000I SQL 命令成功完成。

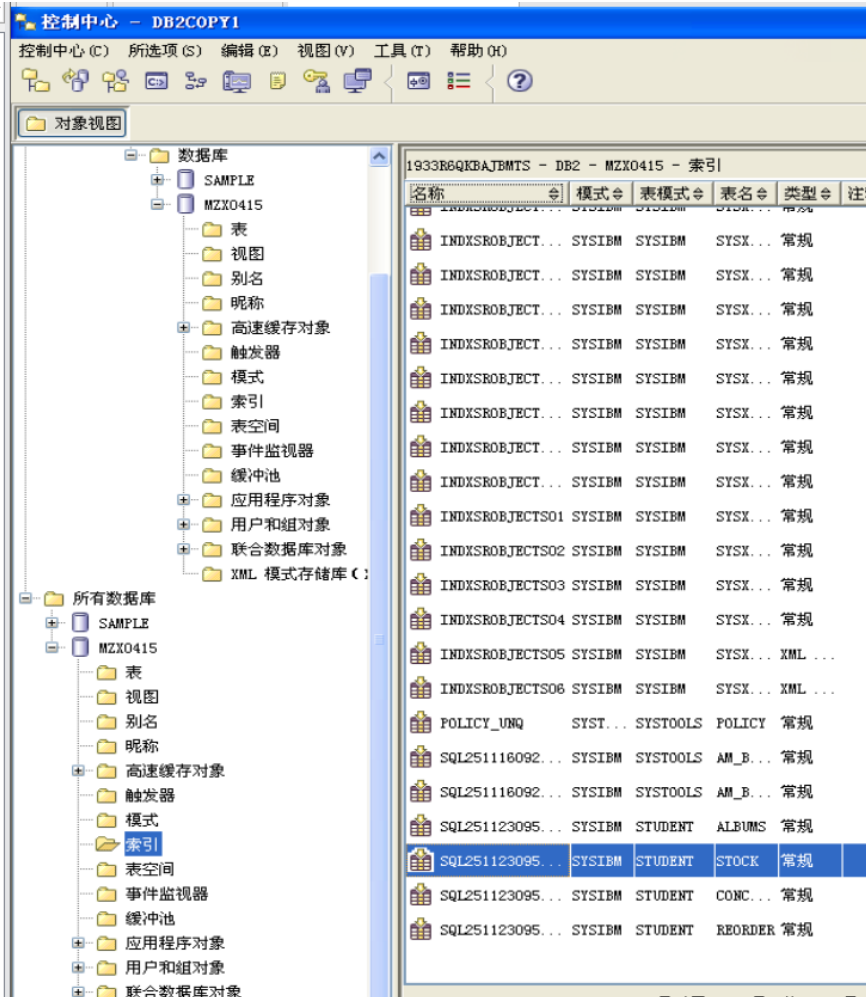
grant select on table reorder to public
DB20000I SQL 命令成功完成。

语句终止字符 :
```

Section 2 - Create Indexes

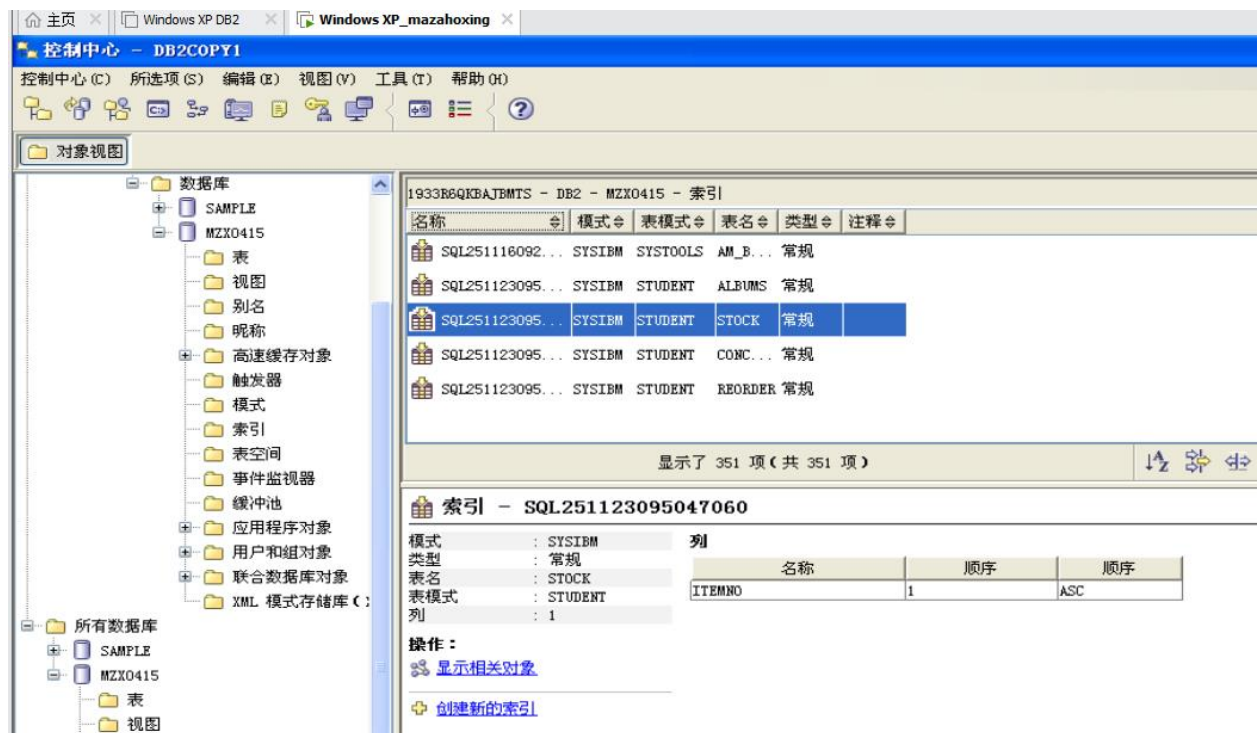
1. Create an index called ITEM on the ITEMNO column in STOCK table.

Show your work below by printing screen.



2. Create a unique index called ITEMNO on the ITEMNO column in ALBUMS table.

Show your work below by printing screen.



3. Select the information from the catalog tables about your indexes. The columns you should select are the first 18 characters of TABNAME, UNIQUERULE, the first 18 characters of INDNAME, and the first 30 characters of COLNAMES from SYSCAT.INDEXES, where INDSHEMA is student, order by TABNAME and INDNAME.

If UNIQUERULE = "U", then only unique values are allowed.

If UNIQUERULE = "D", then duplicate values are allowed.

If UNIQUERULE = "P", then it is a Primary key.

Show your work and your new indexes below by printing screen.

```
substr(indname,1,18), substr(colnames,1,30) from syscat.indexes where indschema = u
ser and uniquerule <> 'P' order by tablename, indname"
```

1	UNIQUERULE	3	4
ALBUMS	U	ITEMNO	+ITEMNO
STOCK	D	ITEM	+ITEMNO

2 条记录已选择。

4. Use the select you executed in the previous step and route the output to indchk.file. Check your results against indchk.master.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.

```
DB2 CLP - DB2COPY1
C:\Labfiles\CF23>db2 -tf indchk.sql > indchk.file
C:\Labfiles\CF23>type indchk.file
```

UNIQUERULE	3	4
ALBUMS	U	ITEMNO
STOCK	D	ITEM

```

2 条记录已选择。

C:\Labfiles\CF23>fc /w indchk.file indchk.master ! more
正在比较文件 indchk.file 和 INDCHK.MASTER
**** indchk.file
2 条记录已选择。 D      ITEM      +ITEMNO
**** INDCHK.MASTER
STOCK      D      ITEM      +ITEMNO
2 record(s) selected.
****

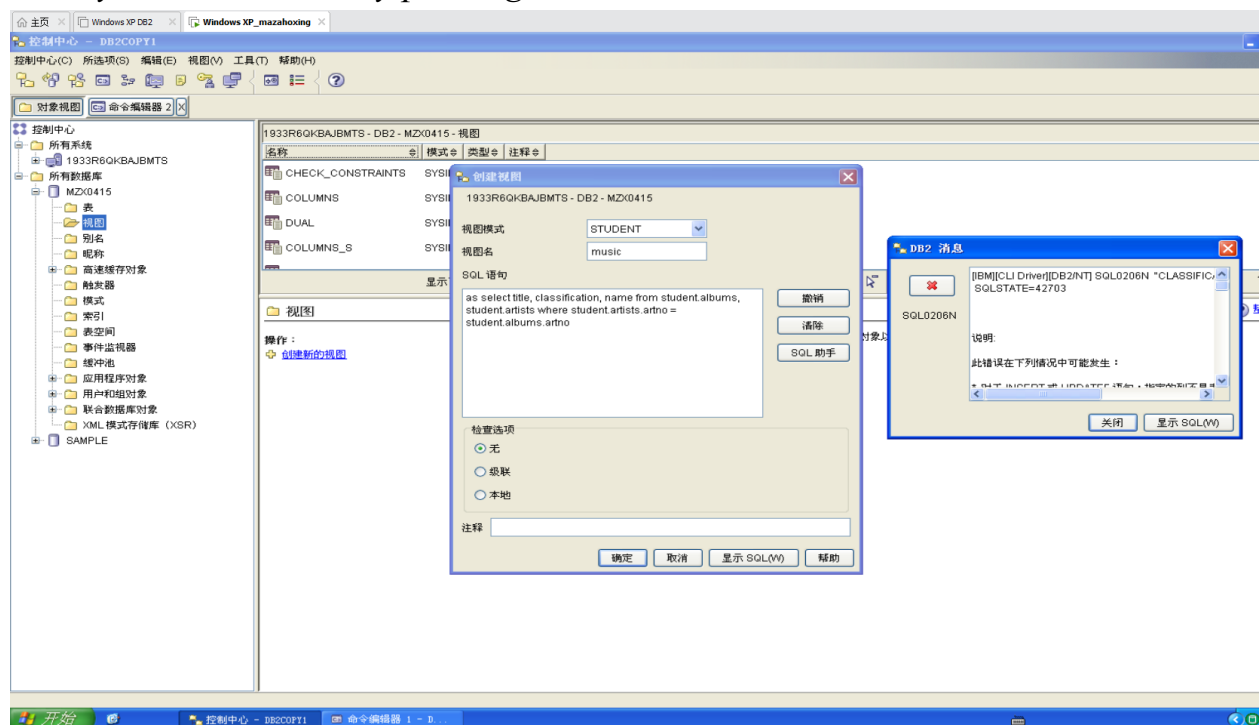
C:\Labfiles\CF23>
```



Section 3 - Create Views

1. Create a view called music that will select title, classification, name from albums, artists where artists.artno = albums.artno.

Show your work below by printing screen.



2. Create another view called inventory by executing the script file, crview.

Show your work below by printing screen.

3. Information about views can be found in the SYSCAT.VIEWS and SYSCAT.TABLES views. Tables and views for the current user can be shown with a LIST TABLES statement. A TYPE of V is a view.

Issue one of the following statements and verify your views are listed.

- *LIST TABLES* statement OR

- *Select columns TABSCHEMA, TABNAME, and TYPE from the SYSCAT.TABLES view*

with a TABSCHEMA of student OR

- *Select columns VIEWSCHEMA, VIEWNAME from SYSCAT.VIEWS with DEFINER of student*

Show your work below by printing screen. 有 bug 完成不了，相同操作最后无法完成

Section 4 - Create Alias

1. Create an alias called singers for the ARTISTS table, and an alias called emptystock for the REORDER table.

Show your work below by printing screen.

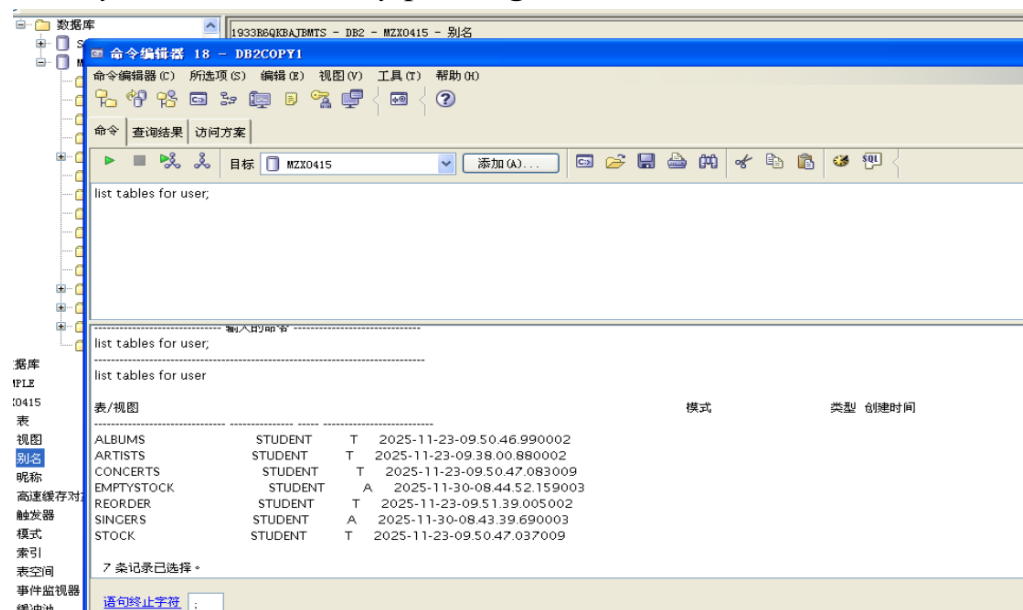


2. Information about aliases can be found in the SYSCAT.TABLES view. Tables, views, and aliases for the current user can also be shown with the LIST TABLES statement. A TYPE of A is an alias. Issue one of the following statements and verify your aliases are listed.

- *LIST TABLES* statement OR

- Select columns TABNAME and TYPE from the SYSCAT.TABLES view with a TABSCHEMA of student.

Show your work below by printing screen.

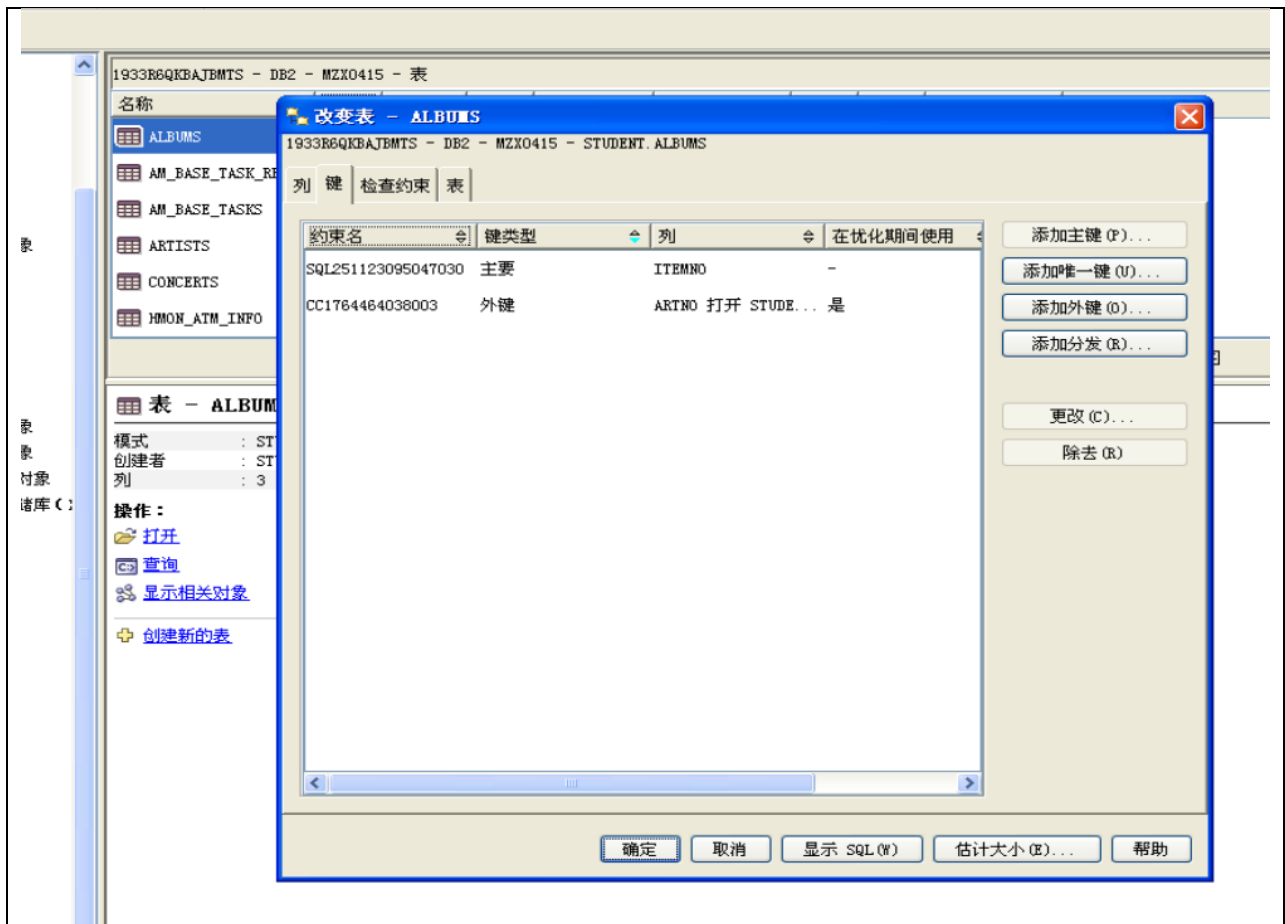


Section 5 - Add Referential Integrity

1. Alter the ALBUMS table and define referential integrity relationships it has with other tables.

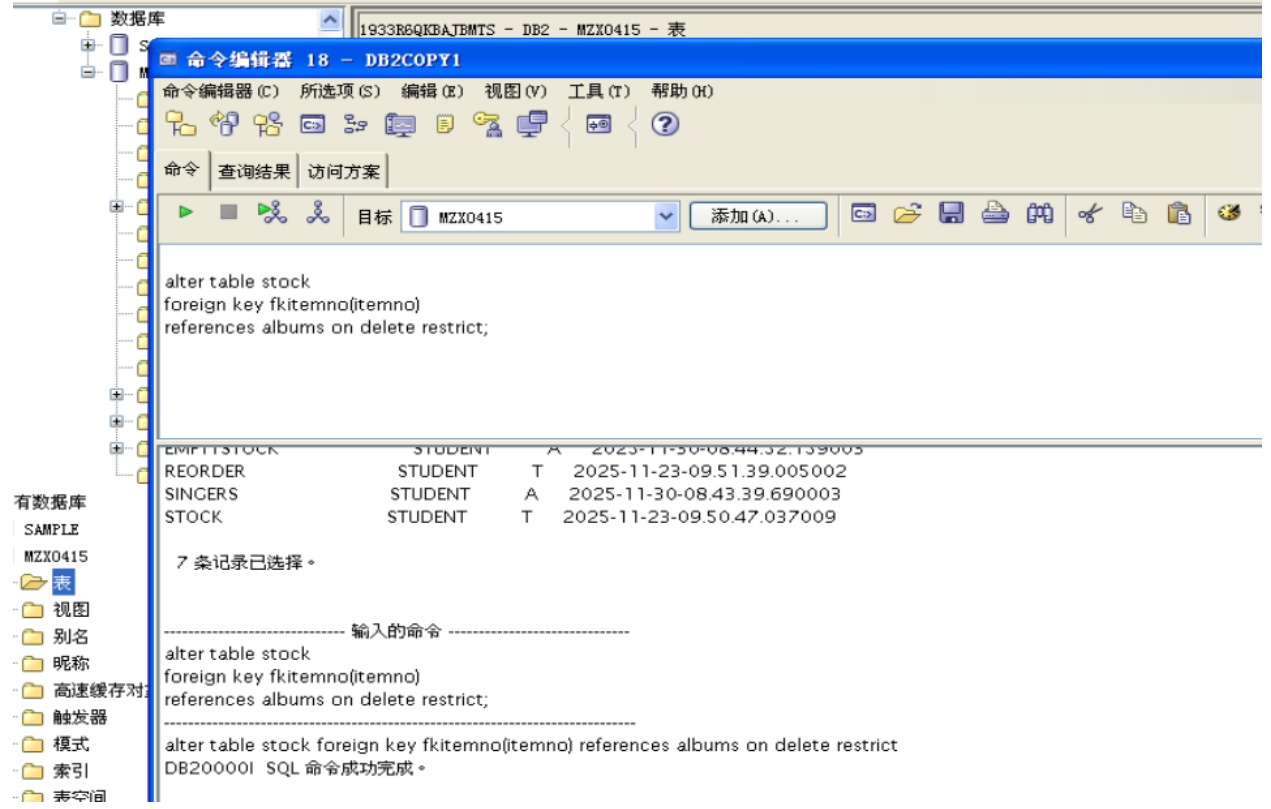
- Add a primary key on the itemno column
- Add a foreign key called fkartno on the artno column which references a primary key in the ARTISTS table
- The delete rule between ARTISTS and ALBUMS should be delete cascade

Show your work below by printing screen.



2. Now alter the STOCK table to define its referential integrity relationships, by executing the script file, crri.

Show your work below by printing screen.



命令编辑器 18 - DB2COPY1

命令 查询结果 访问方案

目标 MZX0415 添加(A)...

```
alter table stock
foreign key fkitemno(itemno)
references albums on delete restrict;
```

EMPTTSTOCK	STUDENT	A	2025-11-30-08.44.32.139003
REORDER	STUDENT	T	2025-11-23-09.51.39.005002
SINGERS	STUDENT	A	2025-11-30-08.43.39.690003
STOCK	STUDENT	T	2025-11-23-09.50.47.037009

7 条记录已选择。

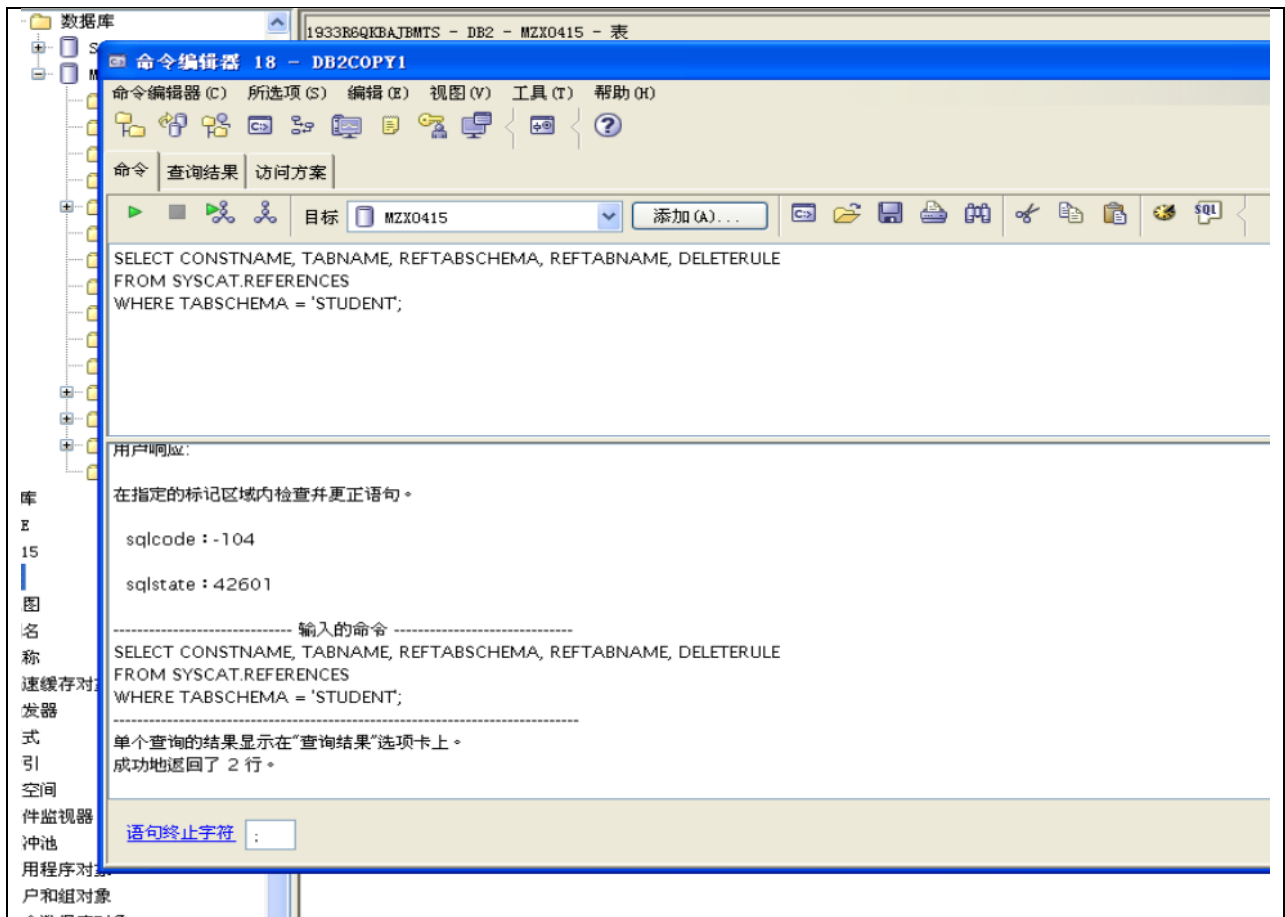
----- 输入的命令 -----

```
alter table stock
foreign key fkitemno(itemno)
references albums on delete restrict;
```

alter table stock foreign key fkitemno(itemno) references albums on delete restrict
DB20000I SQL 命令成功完成。

3. Information about referential integrity can be found by accessing the SYSCAT.REFERENCES view. Table dependencies can also be accessed through the SYSCAT.TABLES view. Enter an interactive SQL statement to select CONSTNAME, TABNAME, REFTABSCHEMA, REFTABNAME, DELETERULE from the SYSCAT.REFERENCES view with a TABSCHEMA of student.

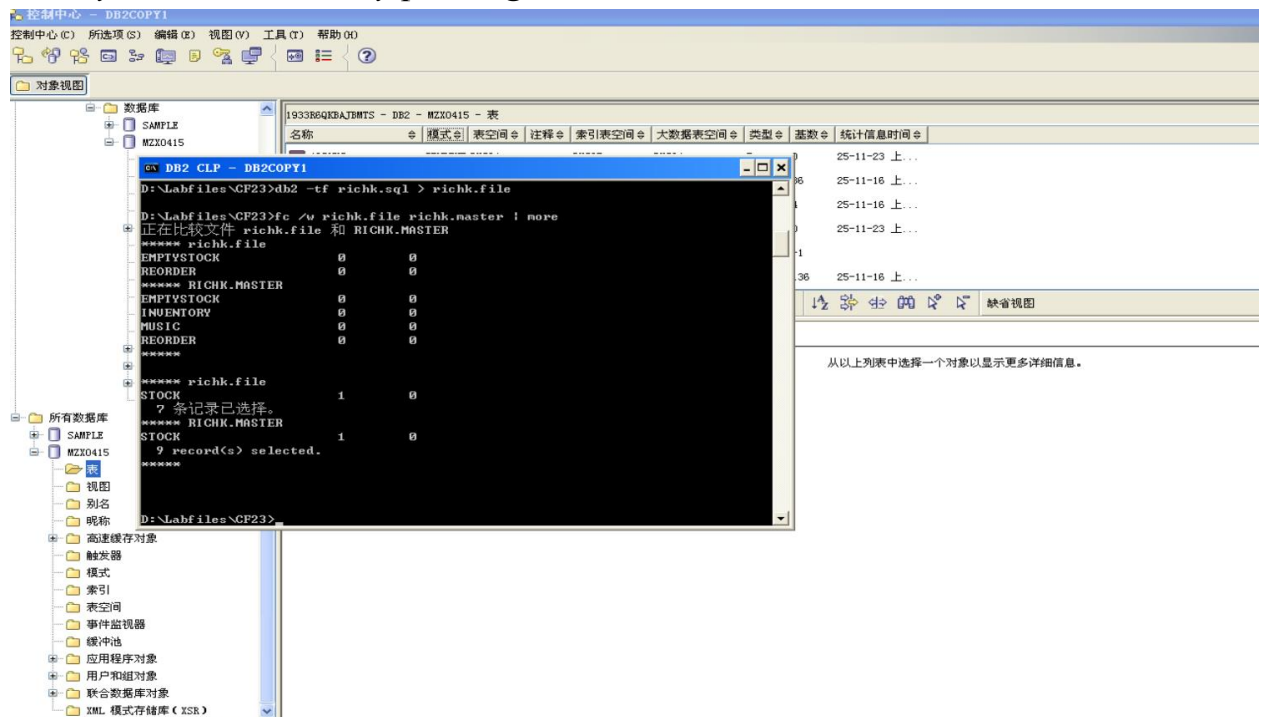
Show your work below by printing screen.



4. Run a select to retrieve the first 18 characters of TABNAME, and the PARENTS and CHILDREN columns from SYSCAT.TABLES where the TABSCHEMA is equal to your userid. Order the results by TABNAME and direct your output to a file named richk.file. The SQL can be found in a file named richk.sql. Check your output against the richk.master file.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.



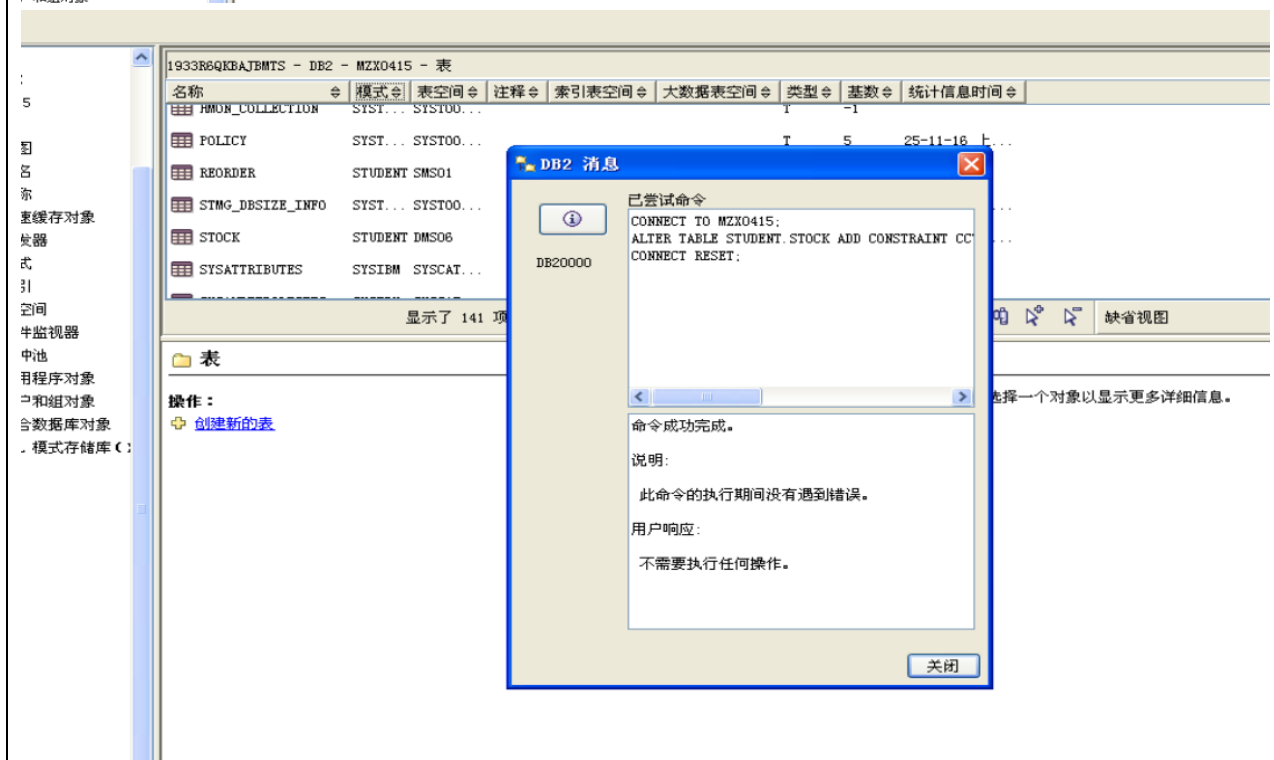
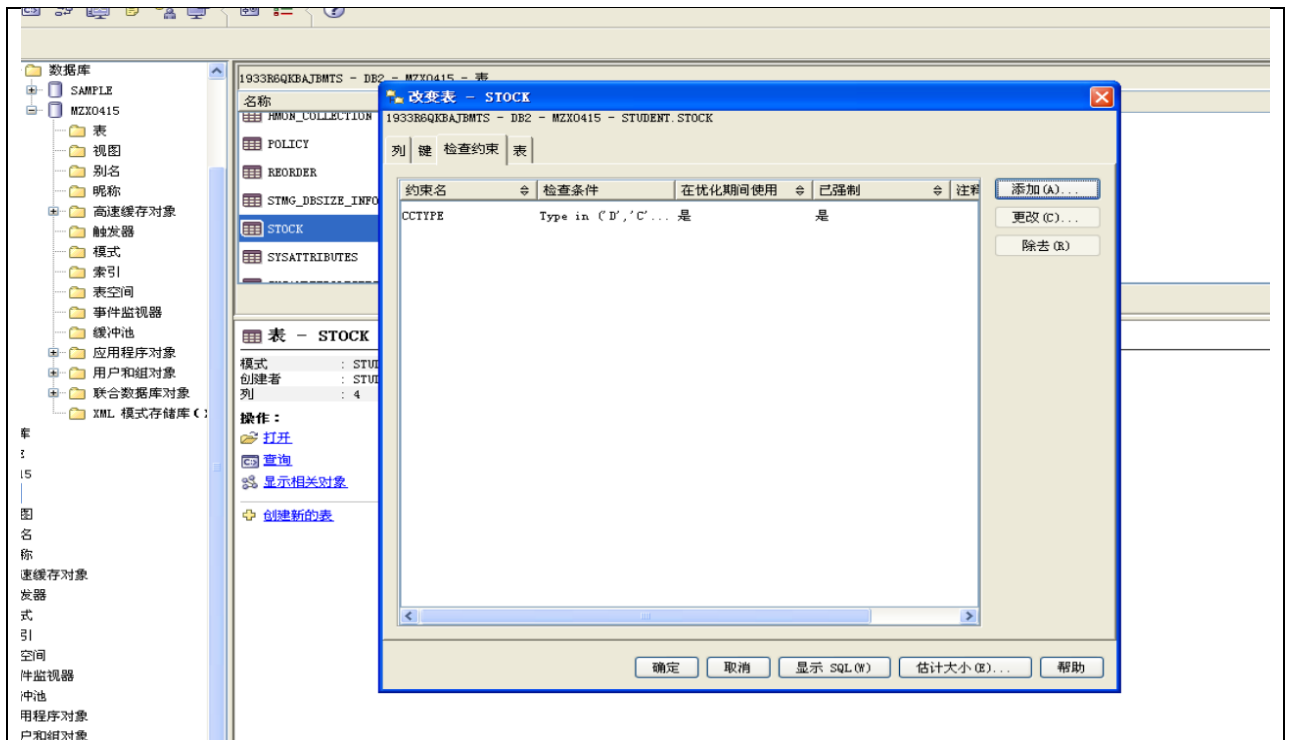
Section 6 - Add Check Constraints

1. Alter the STOCK table and add a check constraint to it. The check constraint should have the following characteristics.

- It should be named cctype.
- The business rule it should enforce is to only allow the values of 'D', 'C', or 'R' in the type column.

Make sure the alphanumeric literals are typed in uppercase letters.

Show your work below by printing screen.



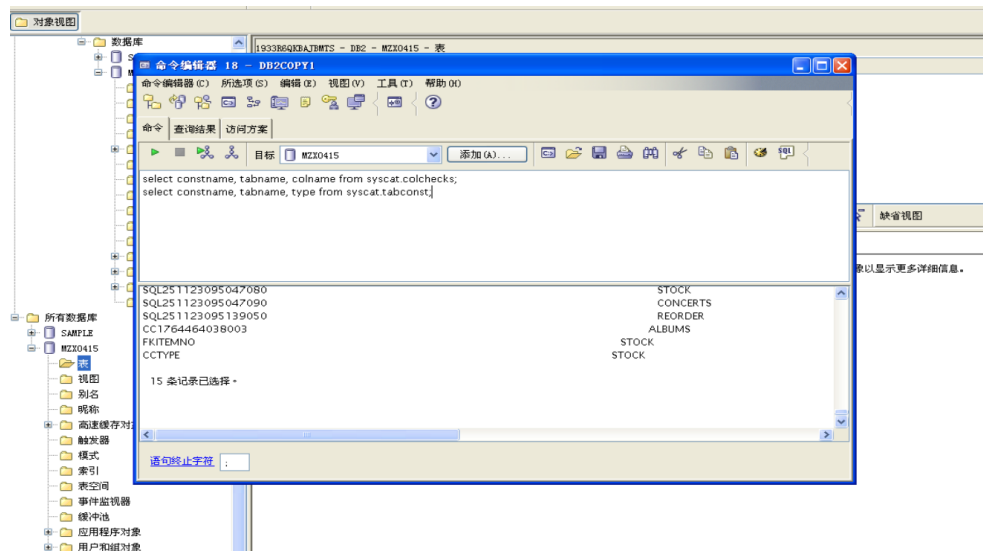
2. Information about check constraints can be found by accessing the SYSCAT.CHECKS, SYSCAT.COLCHECKS, SYSCAT.TABCONST, and SYSCAT.TABLES views. Issue SQL to:

Select CONSTNAME, TABNAME, COLNAME from the SYSCAT.COLCHECKS view.

Select CONSTNAME, TABNAME, TYPE from the SYSCAT.TABCONST view.

If the TYPE = "K", then it is a check constraint. If the Type = "P", then it is a primary key. If the Type = "F", then it is a foreign key.

Show your work below by printing screen.



3. To check your work, run a select statement that selects the first 100 characters of the TEXT column from SYSCAT.CHECKS where CONSTNAME = 'CCTYPE', and direct your output to a file named ckchk.file. The SQL can be found in a file named ckchk.sql. Check your output against the ckchk.master file.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.

```
\Labfiles\CF23>db2 -tf ckchk.sql > ckchk.file

\nLabfiles\CF23>fc /w ckchk.file ckchk.master | more
在比较文件 ckchk.file 和 CKCHK.MASTER
**** ckchk.file
PE in ('D','C','R')

1 条记录已选择。
**** CKCHK.MASTER
PE in ('D','C','R')

1 record(s) selected.
****

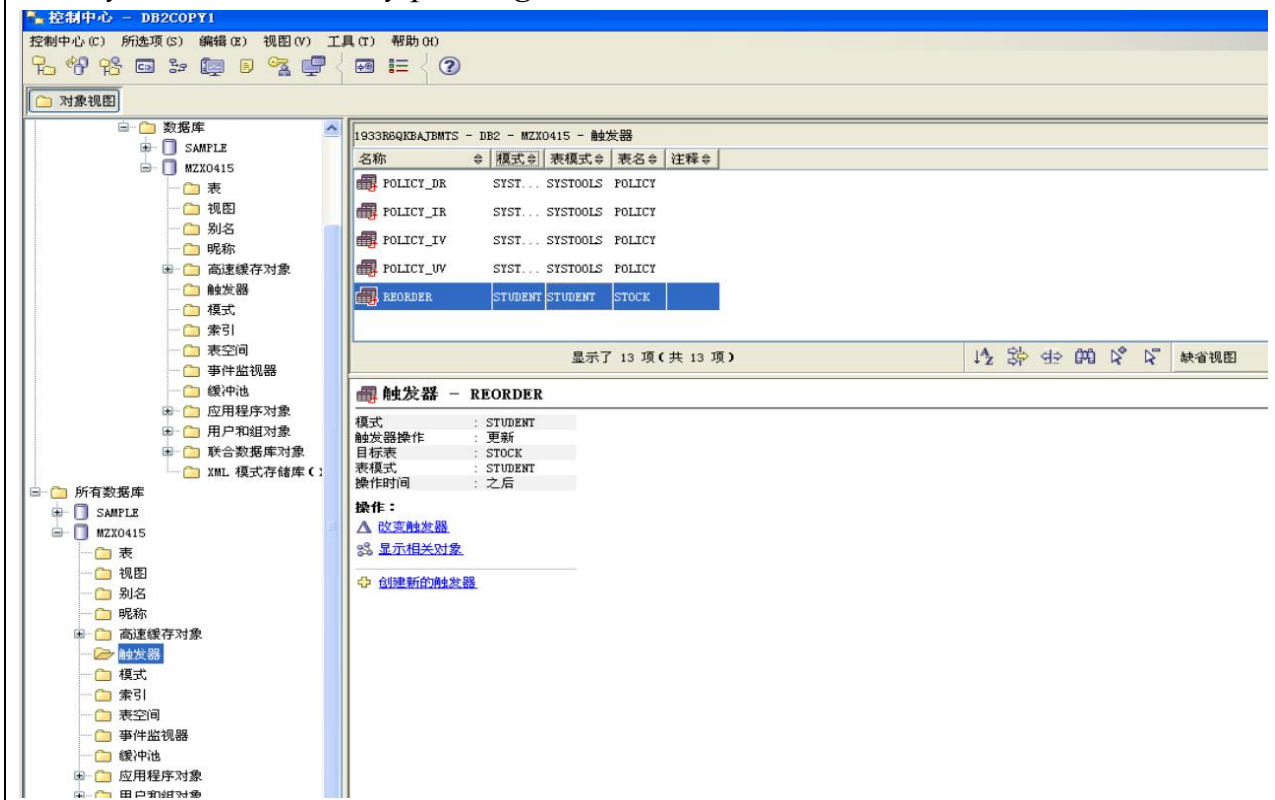
\nLabfiles\CF23>
```

Section 7 - Create a Trigger

1. Create a trigger which has the following characteristics.

- It should be named reorder
- It should fire after an update of the qty column on the STOCK table, if the new value of qty is ≤ 5
- New should be referenced as n
- The triggered action should insert the values n.itemno and current timestamp into the REORDER table
- For each row mode db2sql

Show your work below by printing screen.



2. Information about triggers can be found by accessing the SYSCAT.TRIGGERS and SYSCAT.TRIGDEP views.

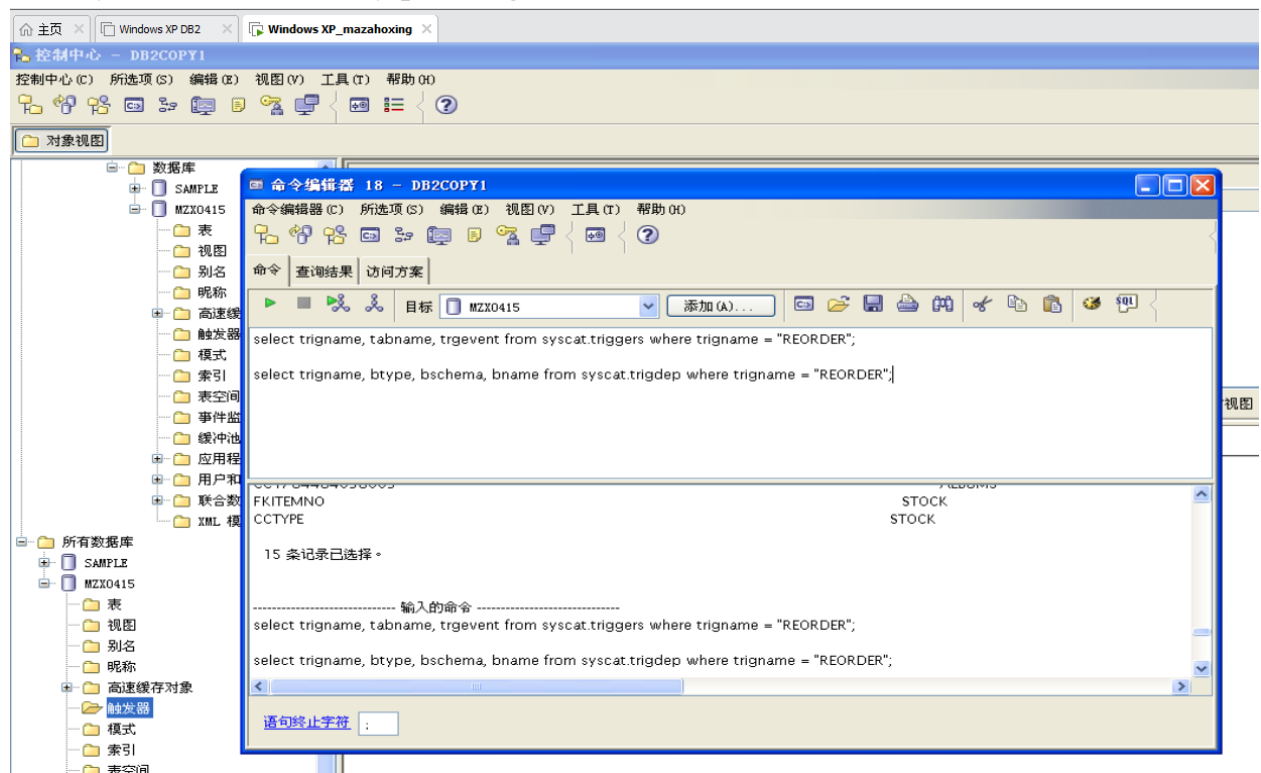
Enter interactive SQL to research the REORDER trigger:

Select TRIGNAME, TABNAME, and TRIGEVEN columns from the SYSCAT.TRIGGERS view.

TRIGEVEN describes the event that fires the trigger as I for insert, D for delete, or U for update.

Select TRIGNAME, BTYPE, BSHEMA, BNAME columns from SYSCAT.TRIGDEP view. BTYPE and BSHEMA give the name of the objected depended on by the trigger. BTYPE describes the type of base object as A for alias, F for function instance, T for table, or V for view.

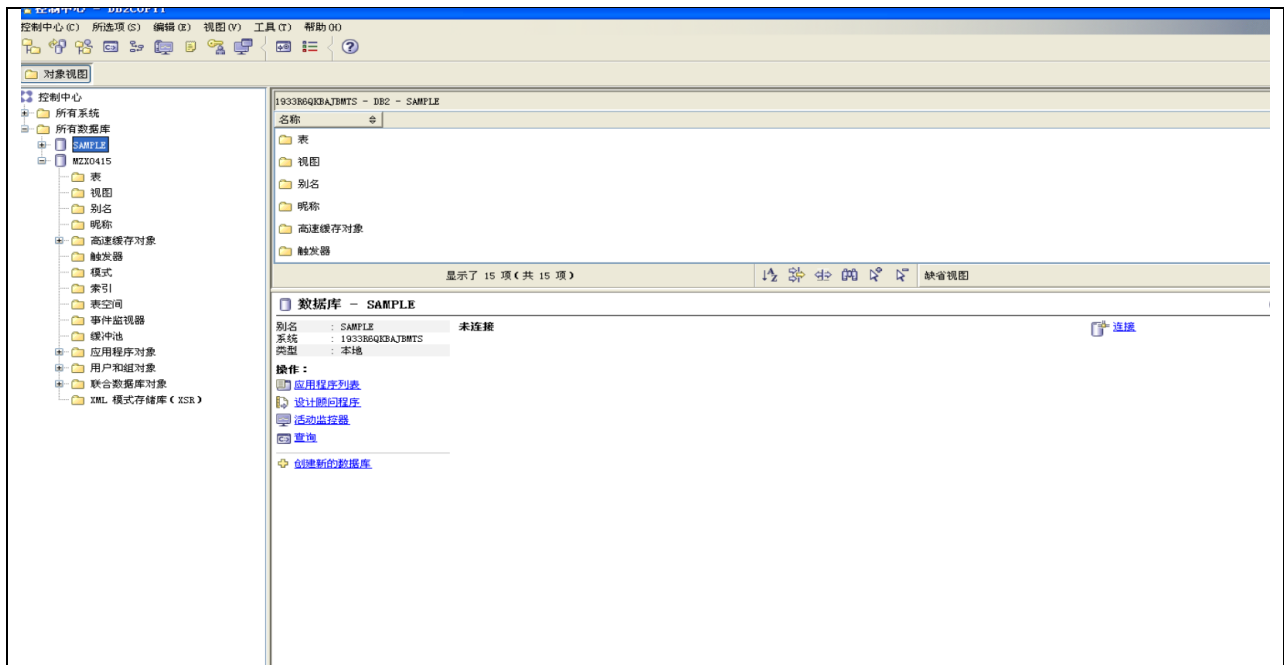
Show your work below by printing screen.



Section 8 - Working with XML

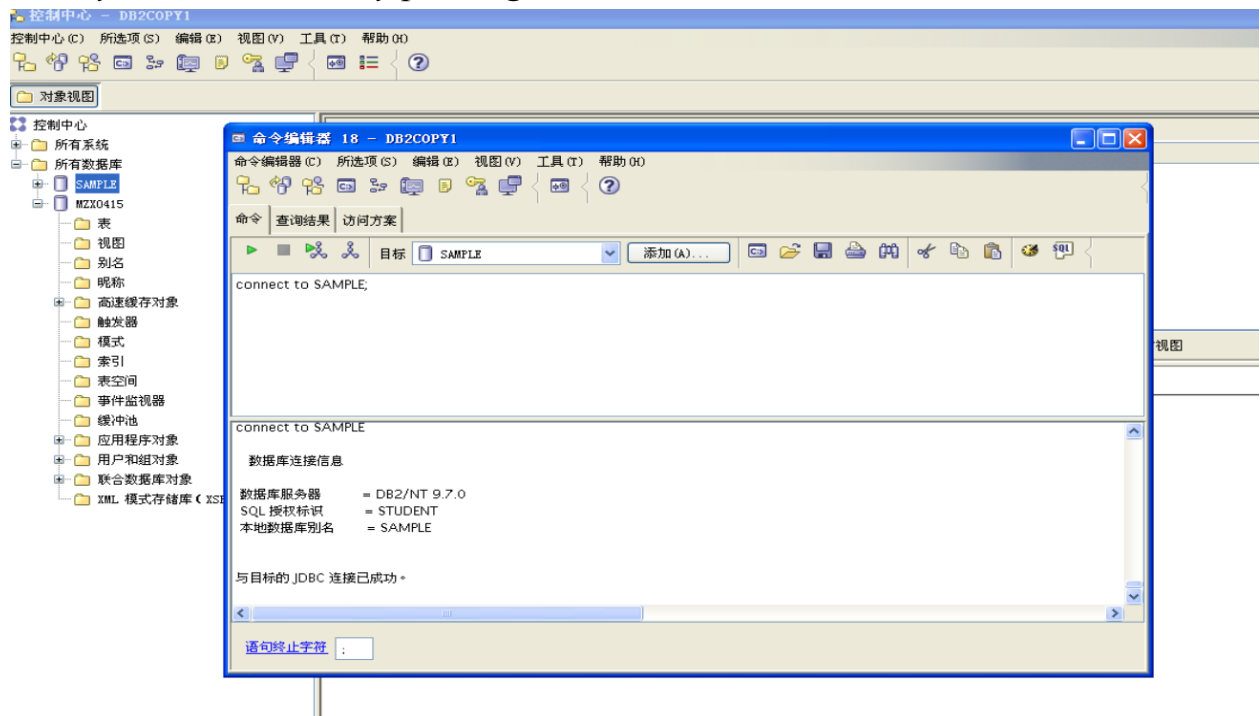
1. If the SAMPLE database has not yet been create, create it now.

Show your work below by printing screen.



2. Connect to the SAMPLE database.

Show your work below by printing screen.



3. Use XQUERY and the db2-fn:xmlcolumn function to retrieve all of the XML documents from the customer table's info column.

Show your work below by printing screen.

The screenshot shows a DB2 Command Editor window titled "命令编辑器 1 - DB2COPY1". The menu bar includes "命令编辑器(C)", "所选项(S)", "编辑(E)", "视图(V)", "工具(T)", and "帮助(H)". The toolbar contains icons for file operations, execution, and help. The "命令" (Command) tab is active, showing the XQUERY statement: `xquery db2-fn:xmlcolumn('ILU.CUSTOMER.INFO');`. The "查询结果" (Query Results) tab is also visible. The result is displayed as XML, showing a customer record for "Goose Defender" with contact information. The status bar at the bottom indicates "6 条记录已选择" (6 records selected) and provides a "语句终止字符" (Statement Termination Character) input field.

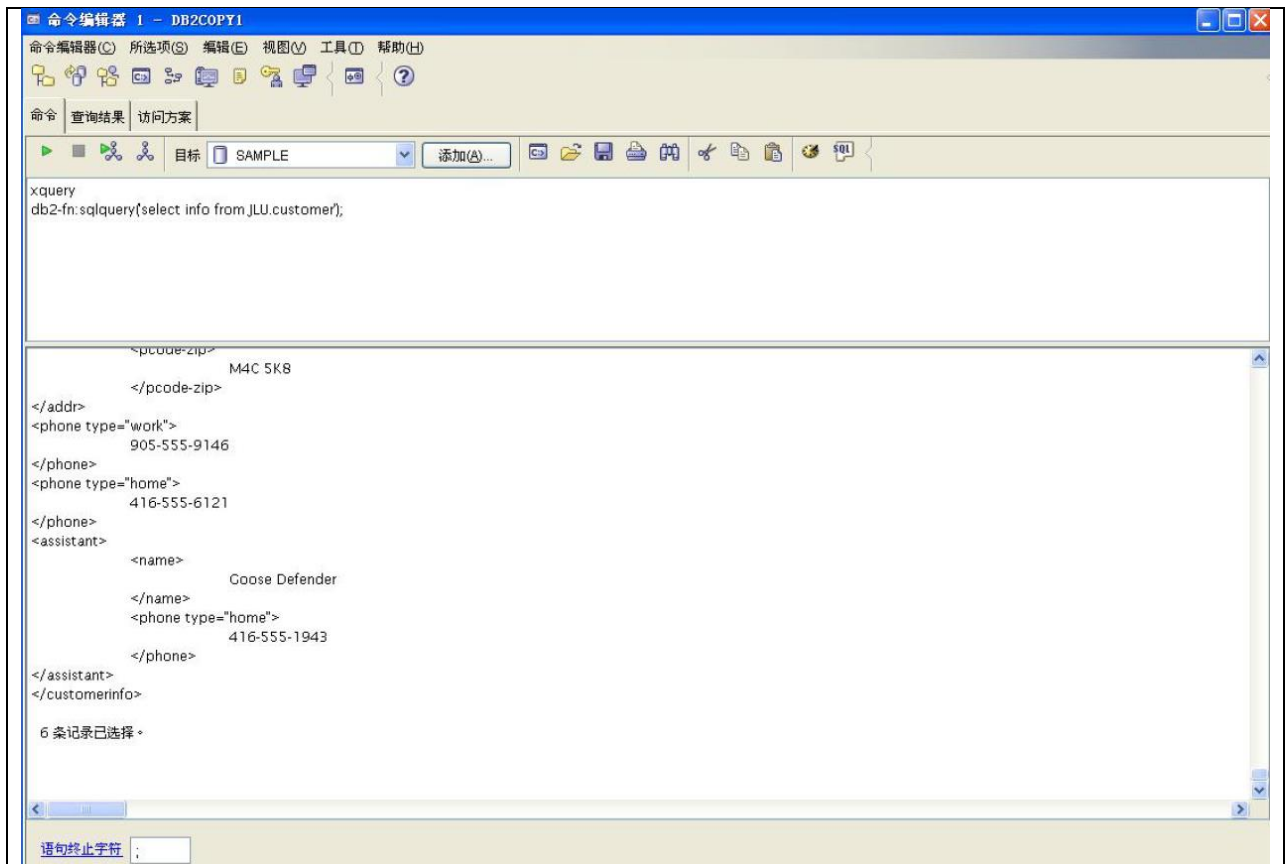
```
<prov-state>
    Ontario
</prov-state>
<pcode-zip>
    M4C 5K8
</pcode-zip>
</addr>
<phone type="work">
    905-555-9146
</phone>
<phone type="home">
    416-555-6121
</phone>
<assistant>
    <name>
        Goose Defender
    </name>
    <phone type="home">
        416-555-1943
    </phone>
</assistant>
</customerinfo>
```

6 条记录已选择 *

语句终止字符:

4. Now use XQUERY with SQL and db2-fn:sqlquery function to retrieve all of the XML documents from the customer table's info column.

Show your work below by printing screen



5. Use XQUERY and the db2-fn:sqlquery function to retrieve, from the INFO column in the CUSTOMER table, all customers where the customer ID (column CID) equals 1002.

Show your work below by printing screen.

